The Gender Gap in Upper-Tail Math Achievement: Evidence from Ukraine

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Abstract

This paper uses Ukraine's national ZNO mathematics exam to study the gender gap at very high achievement levels in a mandatory, near-universal setting. The male to female ratio among top scorers widens steadily and reaches 3.9 to 1 at the 99.9th percentile. The gap appears in almost all schools, with only moderate variation across them, and schools with more girls and stronger overall achievement have a more balanced top tail. Ukraine's International Mathematical Olympiad and European Girls' Mathematical Olympiad teams come from the same small set of elite schools for boys and girls alike, pointing to an extreme-tail gender gap that arises on a conversion margin rather than an access margin.

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1 Introduction

The gender gap in average math test scores among school graduates has narrowed over recent decades to a level of little practical importance (Hyde, Lindberg, et al., 2008; Lindberg et al., 2010; Else-Quest, Hyde, and Linn, 2010). At the upper tail, however, the gap is consistently large: boys are overrepresented among top math performers across countries (Hedges and Nowell, 1995; Machin and Pekkarinen, 2008; Hyde and Mertz, 2009; Breda and Napp, 2018), and at the extreme right tail it widens further, with male-to-female ratios exceeding 10:1 among the very highest achievers (Ellison and Swanson, 2010), a gap already present by ninth grade and widening over the high school years (Ellison and Swanson, 2023). Because this gap is concentrated among the most talented students in math, it may help explain the persistent underrepresentation of women in both STEM education and the STEM workforce. Since STEM fields are among the higher-paying, this underrepresentation also contributes to the gender wage gap (Wiswall and Zafar, 2018; Blau and Kahn, 2017).

This paper examines the upper-tail math gender gap using Ukraine's high-stakes national standardized exam (ZNO), taken by virtually the entire cohort of secondary-school graduates. The exam is administered in roughly 9,500 schools and sat by about 100,000 students each year. Like the American Mathematics Competitions (AMC) studied by Ellison and Swanson (2010), it is far more effective than standard tests at differentiating high-achieving students. I present psychometric evidence that the ZNO math exam has wide score dispersion, difficult items, and strong item discrimination and reliability, even at very high percentiles. The Appendix reproduces several of the hardest questions to illustrate how they continue to separate the strongest students from everyone else.

Ellison and Swanson (2010) note that AMC contests, although well suited to distinguishing among high achievers, are taken only by a self-selected group of students, so voluntary participation raises concerns that their tail estimates partly reflect who chooses to enter contests rather than who actually scores highest. In contrast, ZNO math is not a contest but a

high-stakes standardized exam whose score is used in university admissions, so high-achieving students have little incentive to avoid it. Moreover, mathematics became mandatory in the form of ZNO from 2021 onward, which allows me to observe almost the entire cohort and directly test whether selection into the exam drives the gender gap at the upper tail.

To my knowledge, Ellison and Swanson (2010) remain the leading study to resolve the math achievement tail at this depth using competition microdata. Observing very high percentiles together with school-level information is empirically demanding, and few settings provide such data. Ukraine is one of these settings. The ZNO exam releases individual-level microdata that identify each test-taker's school, which I link to an administrative school census reporting enrollment, staffing, and other characteristics for the universe of schools. Because the exam microdata are universal and school-identified, and the census provides independent information on schools, I can observe both the distribution's upper tail and its correlates at the same time.

In Section 3, I graph the male-female ratio at each percentile of the score distribution. The ratio rises steadily with the threshold, reaching 3.9 to 1 at the 99.9th percentile in 2021, matching the widening pattern in Ellison and Swanson (2010). The gender gap at the extremes of the distribution is also stable across years, including after math became mandatory in 2021, so it does not reflect who chooses to take the exam. This is a new setting for an established finding: the same widening tail gap appears in a mandatory, near-universal exam, where voluntary entry cannot explain it.

Ellison and Swanson (2010) estimate the same beta-binomial model for the AMC 12: at a threshold of 100, roughly the 99th percentile of a general test-taking population, they find a mean female share of 0.18 and a standard deviation of 0.05 across schools, against 0.35 and 0.09 at the 99th percentile of the ZNO (Section 4.1). The higher ZNO mean matches the smaller overall tail gap documented above, but the two standard deviations imply nearly the same coefficient of variation, 0.26 versus 0.28. This supports Ellison and Swanson's main conclusion that cross-school heterogeneity in the gender gap is real but moderate, and shows

it is not an artifact of the AMC's self-selected sample. I reach the same conclusion in a national exam setting: whatever is driving the gender gap is felt broadly across Ukrainian schools. The gap is somewhat larger at some schools and smaller at others, but the differences are modest. Among schools with enough high scorers to make the comparison meaningful, ten or more, 96 percent have more boys than girls among their top scorers.

In Section 4.2, I relate this school-level variation to observable characteristics. Schools with more girls in the exam-taking cohort and higher average scores on the Ukrainian-language exam, my proxy for academic selectivity, have a more balanced top tail; pupil-teacher ratios, spending, income, and urbanization do not. Cohort composition and selectivity, not raw resources, predict where the gap is smaller.

Finally, in Section 5, I turn from cross-school variation to Ukraine's most extreme achievers directly: its International Mathematical Olympiad (IMO) and European Girls' Mathematical Olympiad (EGMO) team members. Ellison and Swanson (2010) find that top U.S. girls come from a narrow, separate set of super-elite schools while top boys are drawn from a much broader pool. Ukraine's pattern is different: IMO and EGMO team members between 2016 and 2021 come from the same 13 schools, 11 of which produce both a boy and a girl at some point, and none of which produce girls only. The concentration of talent at the very top is real, but here it is gender-neutral, which points to a conversion margin rather than an access margin as the source of the gap.

2 The ZNO Mathematics Exam

The Ukrainian External Independent Evaluation (ZNO) is a high-stakes, multi-subject national exam that serves as both a secondary-school graduation requirement and the basis for university admission. This paper focuses on the ZNO mathematics exam, one of the most rigorous components of the test and a required subject for admission to university STEM

programs.[1]

The exam asks students to solve 34 problems in 210 minutes. The problems increase substantially in difficulty over the course of the exam: the first 29 are selected- or short-answer items, while the final five require full written solutions spanning calculus, solid geometry, trigonometry, and parametrized systems of equations, in varying degrees of multi-step problem solving.

In the Ukrainian educational system, achieving a top score on the ZNO math exam carries immense academic prestige. A near-perfect score is the ultimate benchmark of high-school mathematical achievement and serves as the primary gatekeeper for admission into the country's most elite university STEM and quantitative programs. The level of mathematical proficiency required to reach this extreme right tail is exceptionally high; indeed, the students who achieve these top ZNO scores frequently overlap with the highly selected pool of students who represent Ukraine in national and international mathematical olympiads.

Figure 1 plots the distribution of raw ZNO math scores for 2021, for the full population of test-takers. Scores range from 0 to 67 points, with a mean of 19.6 (median 14); 31.1 percent of test-takers score below the passing threshold of 10 points, marked by the dashed line. I define high-achieving students as those scoring above the 99th percentile of the national distribution in the main specification, and turn to the 99.5th and 99.9th percentiles in parts of the analysis that focus on even higher achievers.

For this same population, official psychometrics report a mean item difficulty of 36.7 percent ("difficult"), a discrimination index of 55.9 percent ("very good"), and Cronbach's

[1]Prior to 2021, math was not compulsory: participation among current-year graduates ranged from 47–52% across 2016–2020. In 2021, math became a mandatory graduation requirement, and measured participation rose sharply to 74%.

[2]Count of 2021 ZNO math test-takers by raw score (0–67), one-point bins; $N = 244{,}202$ (including repeaters and prior-year graduates). Dashed line: passing threshold (10 points; 31.1% score below). Dotted lines: 90th, 95th, 99th, and 99.5th percentiles (test-takers above each in parentheses). Source: UCEQA (testportal.gov.ua).

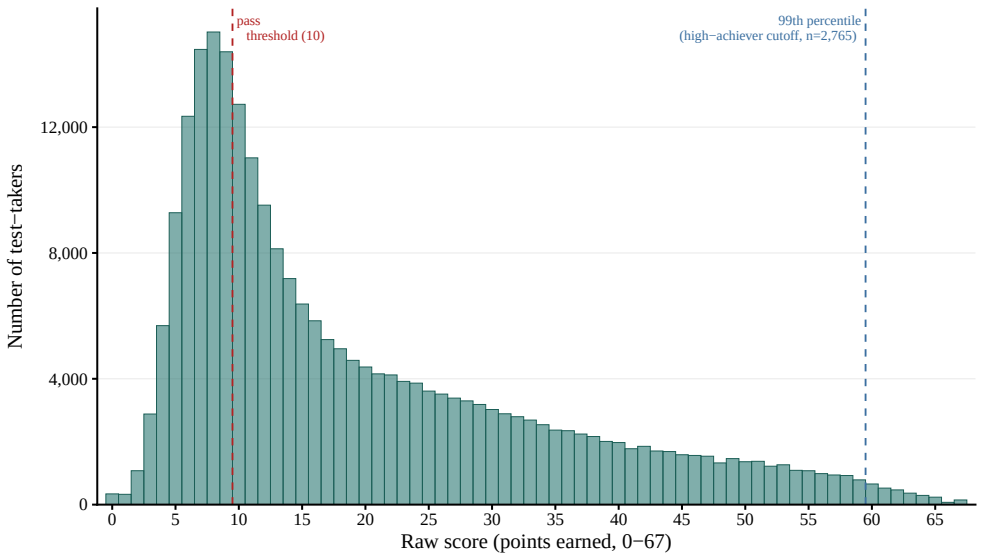


Figure 1: Distribution of Raw Math Exam Scores, 2021[2]

α of 0.94 (Ukrainian Center for Educational Quality Assessment, 2021).[3] A ceiling effect occurs when too many students hit a test's maximum score, so the top bin mixes students of genuinely different ability. Ellison and Swanson (2010) show this for the SAT using matched AMC and SAT scores: students filling the SAT's 99th-percentile bin look, on the harder AMC, more like a mix down to the 97th percentile. The same logic holds for ZNO on its own distribution: only 148 of 244,202 test-takers, 0.061 percent, reached the maximum score of 67, with no comparable bunching at the top—the eight raw scores at or above the 99th percentile (60 through 67, the same 2,765 test-takers shown in Figure 1) each retain a meaningful number of test-takers, ranging from 73 to 656.

[3]Item difficulty (P-value) is the share of test-takers who answer a given item correctly, so a low value means a hard item. Discrimination (D-index) is the gap in success rates on that item between the strongest and weakest scorers overall, so a high value means the item separates students who otherwise do well from those who do not. Reliability (Cronbach's α) is how consistently the test's items agree in ranking the same students, so a high value means the ranking would barely change if a different set of similar questions had been asked.

3 The Gender Gap on the ZNO Math Test

Consistent with Ellison and Swanson (2010), the gender gap in ZNO math scores widens substantially at the upper end of the distribution, though far less steeply than on the AMC: the male-female ratio reaches 3.9-to-1 at the 99.9th percentile in 2021, compared with a 12-to-1 ratio at the same percentile of the AMC 12. Here this pattern shows up in a mandatory, non-competition exam, whose 2021 coverage shift gives a direct test of whether the tail gap reflects selection into testing rather than a genuine ability gap.

Following Ellison and Swanson (2010), let $N_F(\tau)$ and $N_M(\tau)$ denote the number of female and male test-takers scoring at or above cutoff τ. Figure 2 plots the female share, $N_F(\tau)/[N_F(\tau) + N_M(\tau)]$, against the percentile rank of τ, pooling years within each of the two coverage periods.

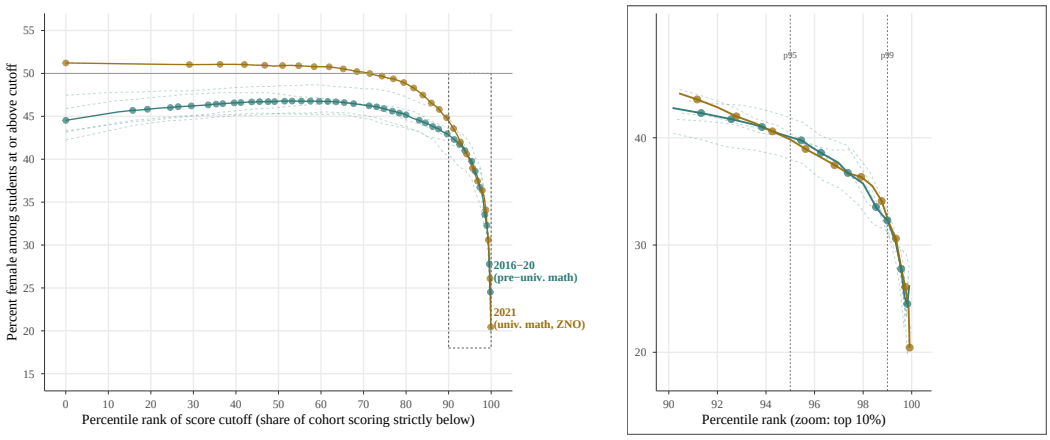


Figure 2: Gender Gaps on the ZNO Math Test, 2016–2020 vs. 2021

Notes: Percent female at or above each percentile of the ZNO math score, pooled by period. Teal: 2016–2020 (optional math). Gold: 2021 (mandatory). Dashed lines: individual years; points: empirical score brackets. Right panel zooms into the top 10 percent of the distribution (90th–100th percentile), with dotted reference lines at the 95th and 99th percentiles; the dashed box in the left panel marks the zoomed region. Source: UCEQA (testportal.gov.ua).

First, the left-most point of each curve gives the overall female share among test-takers. During the voluntary period, 44 percent of test-takers were female, somewhat below parity. Once math became mandatory in 2021, overall female participation rose to 51 percent, essentially reaching parity: pulling in the students who had previously opted out of a voluntary

exam was enough to close the aggregate gap on its own.

Second, the percent female falls sharply in the tail, far below the aggregate figures above. It reaches 32 percent at the 99th percentile of the pooled 2016–2020 distribution, and 26 percent, a 2.8-to-1 male-female ratio, at the 99.9th percentile. The 2021 distribution shows essentially the same shape: 32 percent at the 99th percentile, falling to 20 percent, a 3.9-to-1 ratio, at the 99.9th percentile. The tail gap is thus far larger than the aggregate parity figures would suggest, and it keeps widening well past the 99th percentile in both periods.

Taken together, these two facts speak to a selection concern: before 2021, math was optional, so if unusually able or motivated boys were more likely to opt in than comparably able girls, the tail gap could partly reflect who chooses to test rather than a genuine ability gap. The 2021 shift to a near-universal requirement gives a direct test, since forcing entry should have narrowed the gap if selective participation were driving it. Instead, the tail gap at the 99th percentile is essentially identical across the two periods, even as the aggregate gap closes, so selection into the exam does not appear to be driving the observed tail disparity.

3.1 Beyond the Score Ceiling: Ukranian Math Olympiad

I match 30 of the 34 participants in the 2021 National Math Olympiad grade-11 roster to a confirmed ZNO score. Twenty-six of them, 87 percent, score in the top 1 percent of that year's ZNO math distribution, and eleven score at the exam's maximum of 200. Because ZNO cannot distinguish among students at this ceiling, I use each competitor's own Olympiad score, unbounded at this depth, to examine whether the gender gap continues to widen beyond where ZNO stops resolving differences. Table 1 reports the result.

Panel A of Table 1 shows the ceiling directly: the male-to-female ratio rises from 1.32 at the 90th percentile to 3.01 at the point where the score distribution is exhausted, and cannot rise further. Panel B repeats the exercise among Olympiad participants themselves. Even at the shallowest cut, all 207 participants, the ratio is 4.45, already above ZNO's terminal value. Moving deeper into the Olympiad's own score distribution, the ratio continues to

Table 1: Beyond the ZNO Ceiling: Olympiad Score in the Deepest Tail, 2016–2021

Threshold	N	M:F Ratio	95% CI
Panel A: ZNO Mathematics (N = 643,977)			
Top 10%	64,214	1.32	[1.30, 1.34]
Top 5%	34,374	1.48	[1.45, 1.52]
Top 1%	6,327	2.10	[1.99, 2.22]
Top 0.5%	2,794	2.60	[2.39, 2.82]
Top 0.1%[a]	613	3.01	[2.50, 3.61]
Panel B: National Math Olympiad, Grade 11 (N = 207)[c]			
All participants	207	4.45	[3.13, 6.31]
By score percentile			
Top 50%	111	5.53	[3.32, 9.22]
Top 25%	54	9.80	[4.02, 23.87]
Top 10%	21	9.50	[2.46, 36.71]
Top 5%[b]	11	10.00	[1.65, 60.61]
By competition outcome			
Participant (no prize)	119	3.76	[2.43, 5.82]
Prize (Degree I–III)	88	5.77	[3.23, 10.31]

Notes: Pooled 2016–2021, the years ZNO and the Math Olympiad (UMO) overlap. Panel A restricts to current-year graduates aged 16–20. M:F ratio is males per female at or above each threshold; 95% CIs use the Wilson score interval on the female share. Panel B's score percentiles are defined on each competitor's total competition score; the competition-outcome split combines Degree I, II, and III diplomas into a single Prize category. Source: UCEQA (ZNO), UMO team rosters.

[a] Top 0.05% and 0.01% match this row: the threshold has reached ZNO's maximum score of 200.

[b] Top 1% (N=3, 0 female) omitted

[c] "Grade 11" is the grade-11 cohort proper, excluding composite-grade entrants who sat the grade-11 paper while enrolled in a lower grade (grade labels "11 (10)" and "11 (9)"; $N = 32$). Including them gives $N = 239$ and an all-participant M:F of 5.13.

climb, reaching 9.5 to 1 among the top 10 percent and 10 to 1 among the top 5 percent. The same pattern holds when depth is measured by competition outcome instead of score: participants awarded no prize show a 3.76 to 1 ratio, compared with 5.77 to 1 among those awarded a Degree I–III prize. Whichever measure of depth is used, the gender gap continues to widen well past the point where ZNO itself goes flat.

4 Variation across Schools

In this section I analyze how the gender gap varies across schools, which is potentially important for suggesting policies that might narrow the gap.

4.1 Cross-School Heterogeneity in the Gender Gap

To do this I use no information about the schools themselves, only the number of high-scoring girls and boys each one produces. The strategy is to compare the observed spread of these counts with the spread that chance alone would generate. If all schools were alike, so that a high scorer were female with the same probability everywhere, the girls at a given school would amount to a fixed set of independent draws, and the counts would differ across schools only as sampling noise: a few schools with two or three, many with none, but no more scatter than that noise permits. Genuine differences between schools would instead reveal themselves as *over-dispersion*: counts more spread out than sampling noise allows, with a surplus of schools at zero and a handful producing high-scoring girls repeatedly. Detecting variation in the gender gap therefore reduces to a single question: are the counts more dispersed than a common, school-invariant probability could produce? Making that comparison precise requires an explicit model of the counts, which I now describe.

The model adapts the "dartboard" approach of Ellison and Glaeser (1997), applied to the gender gap by Ellison and Swanson (2010). I treat the number of high-scoring girls at school i, f_i, as binomial, $f_i \sim \text{Binomial}(N_i, p_i)$, where N_i is the school's total number of high scorers

and p_i is the probability that a given high scorer from school i is female. The parameter p_i summarizes how the school's environment bears on the gender gap. The purely random benchmark holds p_i fixed across schools; the alternative is that p_i varies, larger where a school has built an environment that narrows the gap, smaller elsewhere. To test the constant-p model I follow Ellison and Swanson (2010) in assuming the p_i are independent draws from a Beta(α, β) distribution and estimate (α, β) by maximum likelihood. I estimate the mean and variance of p_i this way using the number of girls and boys among the top-percentile scorers in each school on the pooled 2016–2021 mathematics exam.

Table 2 reports the estimated mean and standard deviation of the p_i at the 99th percentile (score ≥ 195) and the 99.5th percentile (score ≥ 197). At the 99th percentile, the p_i are drawn from a distribution with mean 0.35 and standard deviation 0.09. The standard deviation is statistically distinguishable from zero, so the constant-p benchmark is rejected: there are genuinely some schools where the gender gap among high scorers is smaller and others where it is larger. The magnitude, however, is modest. The 99.5th percentile gives the same picture: mean 0.32, standard deviation 0.08. A likelihood-ratio test does not reject equality of the parameters across the pooled years at either threshold, so pooling is justified.

The gap is widespread. Most schools show a male advantage among high scorers, and only a small minority reach parity. The observed variation is larger than in Ellison and Swanson (2010) for the AMC, but it should be interpreted in light of how few girls reach the top tail. With a median of only one high scorer per school, much of the observed variation reflects sampling noise, so detecting systematic differences requires bringing school characteristics directly into the analysis.

4.2 Determinants of the School-Level Gender Gap

I next examine how the gender gap among high scorers varies across schools. For each school with administrative data, the dependent variable is the share of its top-tail math scorers who are female, measured at the 99th and 99.5th percentiles of the national score distribution.

Table 2: Cross-School Variation in the Gender Gap Among High Scorers

	High-scorer threshold	
	99th	99.5th
	(1)	(2)
Score threshold	≥195	≥197
Mean female share, E(p_i)	0.35	0.32
	[0.33, 0.36]	[0.30, 0.33]
SD across schools, SD(p_i)	0.09	0.08
	[0.07, 0.11]	[0.06, 0.11]
Homogeneity across years (p)	0.64	0.65
Schools	1,559	980
High scorers	7,902	4,292

Notes: Beta-Binomial ML estimates, pooling the 2016–2021 ZNO math exams. High scorers are test-takers at or above the stated percentile; p_i is the probability a high scorer at school i is female. E(p_i) and SD(p_i) are the mean and standard deviation of p_i across schools. Score threshold is the pooled empirical cutoff on the 100–200 ZNO math scale corresponding to each percentile. Brackets: 95% delta-method intervals. *Homogeneity across years* is the p-value of a likelihood-ratio test (df=10) that the parameters are equal across the pooled exams; values above conventional levels indicate pooling is not rejected. Co-educational schools only. Source: UCEQA (testportal.gov.ua).

Table 3 relates this share to school, municipal, and regional characteristics.

Two patterns stand out. First, schools with more girls in the exam-taking cohort also tend to have a higher female share among top scorers, but the relationship is less than proportional. A coefficient of about 0.7 means that a school with a female cohort share 10 percentage points higher is associated with only about a 7 percentage point increase in the female share at the top. At the 99.5th percentile, this pass-through falls to about 0.50, so the relationship becomes even weaker deeper in the tail. Second, schools with higher mean Ukrainian scores, my proxy for academic selectivity, also tend to have a more balanced top tail.

By contrast, the other observed characteristics do not show clear associations. Measures of disadvantage, local income and education, pupil-teacher ratios, books, spending, and urbanization are all imprecisely estimated and not statistically distinguishable from zero. Taken together, the results suggest that observable resources and local socioeconomic conditions explain little of the variation in the female share among top scorers.

Because this is a descriptive analysis, the results should be read as patterns rather than causal effects. Still, they point toward the importance of the upper-achievement pipeline. In particular, girls' representation at the top seems to depend more on cohort composition and academic selectivity than on broad resource differences. That is consistent with the idea that the main constraint is not simply school inputs, but the conversion of girls already present in the cohort into very high achievers. This matches Ellison and Swanson (2023)'s decomposition of the widening U.S. AMC gender gap over high school, where they find that differential rates at which lower-ranked girls and boys advance into the top ranks, a conversion margin in this paper's terms, account for more than the entire widening, while differential dropout and entry play a secondary role.

Table 3: Determinants of the Top-Tail Female Share across Schools

	Dependent variable: Female share among high scorers	
	(1) 99th pct.	(2) 99.5th pct.
Cohort female share	0.7319***	0.5008*
	(0.1927)	(0.2795)
Mean Ukrainian score	0.0068**	0.0104**
	(0.0033)	(0.0051)
Share large family	-0.2789	0.1162
	(0.3008)	(0.4703)
Share free meals	0.0197	0.0352
	(0.0859)	(0.1263)
Share higher educ. (area, %)	0.0015	-0.0020
	(0.0022)	(0.0032)
Students per teacher	0.0003	0.0025
	(0.0072)	(0.0102)
Books per student	0.0011	0.0012
	(0.0012)	(0.0017)
Hromada income (1,000 UAH)	0.0138	0.0265
	(0.0126)	(0.0171)
Hromada urban share	-0.1467	0.0583
	(0.0989)	(0.1463)
Observations	539	267
R^2	0.047	0.044

Notes: Unweighted school-level OLS; high scorers are defined at the 99th (col. 1) and 99.5th (col. 2) percentile of the 2021 ZNO math distribution. *Cohort female share* is the female share of the school's Ukrainian-exam participants; *mean Ukrainian score* proxies academic selectivity. Resource variables are winsorized at the 1st and 99th percentiles. The sample excludes private, state, boarding, and count-outlier magnet schools. Standard errors in parentheses. Source: UCEQA administrative microdata (testportal.gov.ua), ZNZ-1 school census, and State Statistics Service.
$^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

5 Concentration at the Very Top: Evidence from the Math Olympiads

Following Ellison and Swanson (2010)'s comparison of the U.S. International Mathematical Olympiad (IMO) team and the China Girls' Mathematical Olympiad team, I turn to Ukraine's most extreme achievers. The IMO is the premier global mathematics competition for secondary students, with national teams selected through country-level olympiads; Ukraine selects its IMO team through a national mathematics olympiad open to all students regardless of gender. Since 2012, Ukraine has also sent a team to the European Girls' Mathematical Olympiad (EGMO), a parallel international competition restricted to high school girls and drawn from the same national olympiad. The IMO team thus represents the country's top performers overall, while the EGMO team represents its top-performing girls. This lets me ask whether Ukraine's most extreme male and female talents come from the same schools or from different ones, and whether the school patterns Ellison and Swanson (2010) document for the United States carry over to this setting.

For each IMO and EGMO team member between 2016 and 2021, I compute how exceptional their school is in terms of producing top-percentile ZNO mathematics scorers. Intuitively, this measures how often a given school produces students who reach the extreme right tail of the national exam distribution, aside from the team member in question. Table 5 in Section B reports the resulting school percentile ranks for all 44 team members.

The pattern looks nothing like Ellison and Swanson's U.S. data. There, the highest-achieving girls are drawn from a much narrower set of schools than the highest-achieving boys, which points to unequal access to the environments that foster extreme achievement. In Ukraine, both teams draw from the same narrow set of schools: 13 schools produced an IMO or EGMO member over the six years, 11 of them produced both a boy and a girl at some point, and none produced girls only. Three schools alone, led by Kharkiv Physics-Mathematics Lyceum No. 27, account for half of all team members. Because IMO and EGMO

team members are drawn from the same elite schools regardless of gender, the remaining extreme-tail gender gap must arise within a shared school pipeline, in how existing talent is converted into very top scores, rather than from unequal access to the relevant schools.

Earlier sections showed that boys are more likely than girls to reach the extreme right tail of the ZNO distribution: the female share among top scorers falls from about one-third at the 99th percentile to one-fifth at the 99.9th percentile, and the beta-binomial estimates indicate that almost all schools with many high scorers have more boys than girls at the top. The IMO-EGMO evidence adds that this male advantage arises within a common elite pipeline: the same small group of physics-mathematics lyceums produces both extreme boys and extreme girls.

This shared pipeline reflects Ukraine's inherited Soviet system of specialized physics-mathematics schools (fizmat), designed to concentrate and intensively train high-ability students in mathematics and science (Gerovitch, 2019; Karp and Vogeli, 2013). The schools that produce IMO and EGMO team members are overwhelmingly these fizmat lyceums, so Ukraine's extreme boys and girls are drawn from the same highly selective math track rather than from separate school systems.

6 Conclusion

This paper documents four facts about the gender gap in upper tail math achievement in Ukraine. First, the male to female ratio among high scorers on the ZNO widens steadily with the percentile threshold and reaches 3.9 to 1 at the 99.9th percentile in 2021. This mirrors the widening pattern that Ellison and Swanson (2010) find for the AMC, but here in a mandatory national exam, where voluntary entry cannot explain the tail gap. Second, a beta binomial model of cross school heterogeneity shows that the gap is present almost everywhere. Among schools with enough high scorers to compare, the large majority have more boys than girls at the top, but the dispersion in the size of the gap across schools

is real and modest rather than driven by a few outliers. Third, the school characteristics that are associated with a larger female share among top scorers are cohort composition and academic selectivity rather than raw resources. Schools with more girls in the exam taking cohort and higher average scores on the Ukrainian language exam have a more balanced top tail, while pupil teacher ratios, spending, and local income do not. Fourth, Ukraine's most extreme achievers, its IMO and EGMO team members, are drawn from the same narrow set of elite schools regardless of gender, unlike the separate, narrower pipeline that Ellison and Swanson (2010) document for U.S. girls.

Taken together, these results locate the gap on a conversion margin rather than an access margin. Girls are present in the relevant schools and cohorts in the numbers one would expect. What differs is the rate at which that presence converts into a top score. Ellison and Swanson (2023) reach a similar conclusion in a very different setting. Using AMC panel data, they show that differential rates at which students move up into the very top ranks account for more than the entire widening of the U.S. tail gap, while differences in entry and dropout play a more limited role. The ZNO is a single sitting exit exam rather than a repeated competition, so I cannot observe comparable individual trajectories, but the cross sectional evidence points to the same margin. Why fewer girls convert into extreme achievers remains an open question. The two papers this project builds on suggest several possible explanations without resolving among them. Ellison and Swanson (2010) propose that girls may be more compliant with standard coursework and less likely to seek out the accommodations or extracurricular opportunities needed to reach the very top, or that the female ability distribution simply has less variance in the extreme right tail. They also note that a lower number of extreme high achieving girls need not reflect a deficiency if it instead reflects girls choosing broader skill portfolios. Ellison and Swanson (2023) add that a lower valuation of competitive math achievement, a weaker taste for competition itself, or a larger reaction to falling just short of a qualifying threshold could each generate a persistent conversion gap. These explanations are not mutually exclusive, and nothing in this paper

distinguishes between them.

Further work on this project aims to bring the ZNO setting to bear on these questions. A national exam that covers multiple subjects offers a margin that contest data do not. It allows me to test whether girls who fall short at the very top of math nonetheless outperform in other subjects, which would speak directly to a portfolio choice story. The next step is to use this margin to ask not only where the gap sits, but why.

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Appendix A Cross-School Variation, 2021 Only

Table 2 in the main text pools the 2016–2021 exams to estimate the cross-school heterogeneity distribution as precisely as possible, which it can because the model uses only gender counts. The school-level covariate regression, by contrast, is restricted to 2021, the year for which the school characteristics are measured. Table 4 re-estimates the beta-binomial on that same 2021-only sample, the 592 coeducational schools with at least one high scorer at the 99th percentile, so the heterogeneity estimates line up with the sample used in the regression. The point estimates are close to the pooled ones: at the 99th percentile, $\mathrm{E}(p_i) = 0.33$ and $\mathrm{SD}(p_i) = 0.10$, against 0.35 and 0.09 when pooled.

Table 4: Cross-School Variation in the Gender Gap, 2021 Only

	High-scorer threshold		
	99th	99.5th	99.9th
	(1)	(2)	(3)
Mean female share, $\mathrm{E}(p_i)$	0.33	0.29	0.27
	(0.01)	(0.02)	(0.04)
SD across schools, $\mathrm{SD}(p_i)$	0.10	0.09	0.16
	(0.03)	(0.04)	(0.06)
Schools	592	308	109
High scorers	1,512	707	208

Notes: Beta-Binomial ML estimates on the 2021 ZNO math exam (the school-regression sample). High scorers are test-takers at or above the stated percentile; p_i is the probability a high scorer at school i is female. Delta-method standard errors in parentheses. The 99.9th-percentile column (109 schools, median one high scorer each) is underpowered; its $\mathrm{SD}(p_i)$ is uninformative. Coeducational schools only. Source: UCEQA (testportal.gov.ua).

Appendix B IMO and EGMO Team Members' School Percentile Ranks, 2016–2021

Table 5 reports the leave-one-out school concentration measure discussed in Section 5 for every Ukrainian IMO and EGMO team member, 2016–2021. For each student, "Cls." is the number of schoolmates who also scored at or above the stated ZNO math percentile that year, excluding the student's own score, and "School %ile" ranks that count against every Ukrainian school with a math test-taker that year, following Ellison and Swanson (2010)'s Table 5 leave-one-out procedure.

Table 5: IMO and EGMO Team Members: School Percentile Ranks, 2016–2021

			99th pctile		99.5th pctile	
Year	Contestant	School	Cls.	School %ile	Cls.	School %ile
Panel A: IMO team members						
2021	Andrii Kovryhin	Lyceum #208	10	99.7	7	99.8
2021	Ihor Pylaiev	Kharkiv Physics-Math Lyceum #27	27	99.9	15	99.9
2021	Klymentii Zolotarov	Kharkiv Physics-Math Lyceum #27	27	99.9	15	99.9
2021	Mykyta Kharin	Mariupol Technical Lyceum	16	99.8	8	99.8
2021	Sviatoslav Denyskov	Kharkiv Physics-Math Lyceum #27	27	99.9	15	99.9
2021	Zahar Naumets	Kyiv-Pechersk Lyceum #171 “Leader”	23	99.9	17	99.9
2020	Fedir Yudin	Lyceum #208	6	99.7	6	99.7
2020	Marko Khasin	Rusanivskyi Lyceum	10	99.8	9	99.9
2020	Oleksii Masalitin	Novopecherska School	2	98.9	2	99.2
2020	Yurii Hladkov	Kharkiv Physics-Math Lyceum #27	15	99.9	15	99.9
2019	Oleksandr Voitovych	Kharkiv Physics-Math Lyceum #27	17	99.9	10	99.9
2019	Oles Holembovskyy	Uzhhorod Specialized Boarding School	9	99.8	4	99.7
2019	Vadym Koval	UPML (Taras Shevchenko Natl. Univ.)	33	99.9	22	99.9
2019	Volodymyr Fedynyak	Lviv PML-B (Ivan Franko Natl. Univ.)	36	99.9	23	99.9
2018	Alina Harbuzova	Kharkiv Physics-Math Lyceum #27	29	99.9	17	99.9
2018	Arsenii Nikolaiev	Kyiv Natural Science Lyceum #145	27	99.9	17	99.9
2018	Illya Koval	Kharkiv Academic Lyceum #45	14	99.9	8	99.9

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Table 5 continued from previous page

			99th pctile		99.5th pctile	
Year	Contestant	School	Cls.	School %ile	Cls.	School %ile
2018	Nhok Tkhai Shon Nho	Kharkiv Physics-Math Lyceum #27	29	99.9	17	99.9
2018	Roman Sarapin	Kharkiv Physics-Math Lyceum #27	29	99.9	17	99.9
2017	Anton Trygub	Gymnasium “Akademiia”	3	99.3	0	0.0–98.0
2017	Heorhii Ivanchyk	Kyiv-Pechersk Lyceum #171 “Leader”	17	99.9	13	99.9
2017	Oleh Rudenko	Kharkiv Physics-Math Lyceum #27	15	99.9	4	99.8
2016	Denys Pushkin	Lyceum “Naukova Zmina”	14	99.9	7	99.9
2016	Nikita Skybytskyi	Lyceum #208 (Dniprovskyi district)	5	99.7	3	99.7
2016	Olha Silina	Kharkiv Physics-Math Lyceum #27	19	99.9	10	99.9
2016	Yaroslava Kivva	Kyiv-Pechersk Lyceum #171 “Leader”	22	99.9	16	99.9
Panel B: EGMO team members						
2021	Karolina Susol	Uzhhorod Specialized Boarding School	10	99.7	7	99.8
2021	Roksolana Ivanchuk	Lyceum “Naukova Zmina”	8	99.7	3	99.5
2021	Sofia Ryabchun	Rusanivskyi Lyceum	17	99.8	12	99.9
2021	Yelyzaveta Horokh	Kyiv Natural Science Lyceum #145	25	99.9	12	99.9
2020	Yana Kolodach	Kyiv-Pechersk Lyceum #171 “Leader”	21	99.9	16	99.9
2020	Yuliia Tatarinova	Kharkiv Physics-Math Lyceum #27	17	99.9	17	99.9
2019	Karyna Nechyporuk	Lyceum #208	13	99.9	5	99.8
2019	Kateryna Horokh	Kyiv Natural Science Lyceum #145	26	99.9	11	99.9

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Table 5 continued from previous page

			99th pctile		99.5th pctile	
Year	Contestant	School	Cls.	School %ile	Cls.	School %ile
2019	Olesia Bilyk	Kyiv Natural Science Lyceum #145	26	99.9	11	99.9
2019	Vlada Petrusenko	Novopecherska School	1	97.2	1	98.6
2018	Alina Harbuzova	Kharkiv Physics-Math Lyceum #27	31	99.9	19	99.9
2018	Anastasiia Rudenko	Kyiv-Pechersk Lyceum #171 "Leader"	19	99.9	8	99.9
2017	Alina Yan	UPML (Taras Shevchenko Natl. Univ.)	30	99.9	15	99.9
2017	Olha Shevchenko	Kharkiv Academic Lyceum #45	4	99.4	2	99.5
2017	Yuliia Zdanovska	UPML (Taras Shevchenko Natl. Univ.)	30	99.9	15	99.9
2016	Oleksandra Tumak	Lviv PML-B (Ivan Franko Natl. Univ.)	31	99.9	16	99.9
2016	Olga Shevchenko	Kharkiv Academic Lyceum #45	13	99.9	10	99.9
2016	Olha Silina	Kharkiv Physics-Math Lyceum #27	20	99.9	11	99.9
Panel C: Summary						
-	*Median, male IMO members*	-	16	99.9	10	99.9
-	*Median, EGMO members*	-	19.5	99.9	11	99.9

Notes: "Cls." is the number of schoolmates who scored at or above the stated percentile on that year's ZNO math exam, excluding the team member's own score (leave-one-out). "School %ile" ranks that count against every Ukrainian school with a math test-taker that year, following Ellison and Swanson (2010)'s Table 5 procedure; when no other school nationally ties the student's leave-one-out count of zero, the percentile is reported as the range spanning all zero-classmate schools rather than a single number, matching Ellison and Swanson's own treatment of this case. Percentiles

rounded to one decimal and capped at 99.9: no school in this sample is literally at the 100th percentile (the true maximum is 99.99), so a value that would round up to 100.0 is instead reported as 99.9 rather than implying literal rank one. Repeat team members (2016–2021) show only their most recent attempt. "UPML" abbreviates Ukrainian Physics-Math Lyceum; "Lviv PML-B" abbreviates Lviv Physics-Math Lyceum-Boarding School. Panel C reports median classmates and median school percentile for male IMO team members (excluding the three female IMO team members in this period) and for EGMO team members, all of whom are female by construction. Source: IMO/EGMO official results and team rosters (imo-official.org, egmo.org), matched by hand to UCEQA ZNO administrative microdata (testportal.gov.ua).

Appendix C Olympiad Entrants' Schools, 2021

Table 6 repeats the Section B exercise one selection tier lower, for every grade-11 participant in the 2021 National Math Olympiad (UMO) rather than only the students who went on to represent Ukraine internationally. Two features make this a cleaner test of Ellison and Swanson (2010)'s question than the IMO/EGMO comparison. First, UMO is a single competition with open entry, so male and female participants clear an identical selection threshold; the IMO/EGMO contrast instead compares an open competition with a girls-only one, whose selection depths need not match. Second, gender is observed directly rather than proxied by which team a student joined.

Table 6: National Math Olympiad Grade-11 Participants: School Strength, 2021

		Top 10% (≥178)		Top 5% (≥186)		Top 1% (≥195)	
Participant	School	Cls.	School %ile	Cls.	School %ile	Cls.	School %ile
Panel A: Male participants							
Danylo Hotskivskyi	Lviv PML-B (Ivan Franko Natl. Univ.)	126	99.9	111	99.9	60	99.9
Pavlo Kryven	Lviv PML-B (Ivan Franko Natl. Univ.)	126	99.9	111	99.9	60	99.9
Maksym Andriichuk	UPML (Taras Shevchenko Natl. Univ.)	82	99.9	70	99.9	39	99.9
Artem Sdobnov	Kharkiv Physics-Math Lyceum #27	49	99.7	46	99.8	29	99.9
Klymentii Zolotarov	Kharkiv Physics-Math Lyceum #27	49	99.7	46	99.8	29	99.9
Andrii Vlasenko	Richelieu Science Lyceum (Odesa)	87	99.9	61	99.9	23	99.9
Zahar Naumets	Kyiv-Pechersk Lyceum #171 "Leader"	42	99.6	35	99.7	23	99.9
Mykyta Kharin	Mariupol Technical Lyceum	69	99.9	43	99.8	16	99.8
Ivan Hechu	Dnipro Oblast Physics-Math Boarding	26	99.2	25	99.6	13	99.8
Volodymyr Didur†	Gymnasium #178 (Solomianskyi, Kyiv)	53	99.8	33	99.7	12	99.8
Andrii Zhuravlov	Zhytomyr City Lyceum #1	48	99.7	35	99.7	11	99.7
Andrii Kovryhin†	Lyceum #208	58	99.8	40	99.8	10	99.7
Anton Uvarenko	Lyceum #35 (Kramatorsk)	13	97.3	8	97.7	5	99.5
Maksym Tur	Poltava Oblast Science Boarding	33	99.4	21	99.4	4	99.2
Valerii Mykhailov	Sievierodonetsk Multi-Profile Lyceum	25	99.1	18	99.3	4	99.2
Ivan Spyrydonov	Kryvyi Rih Gymnasium #95	31	99.4	18	99.3	4	99.2
Arsenii Liulka	Lyceum #35 (Kramatorsk)	13	97.3	8	97.7	4	99.2
Serhii Shevchuk	Rivne Oblast Science Lyceum	56	99.8	29	99.7	3	98.9
Yehor Hudenko	Mykolaiv Municipal Collegium	22	98.9	15	99.2	2	98.1
Andrii Pillei	Physics-Math Gymnasium #17 (Vinnytsia)	23	99	12	98.8	2	98.1
Volodymyr Kovalov	Mykolaiv Specialized School #22	13	97.3	8	97.7	1	93–97.5
Daniil Zabilskyi	Mariupol City Lyceum	18	98.4	7	97.3	1	93–97.5
Denys Yanytskyi	Ternopil Technical Lyceum	10	96.1	7	97.3	0	0–93.4

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Table 6 continued from previous page

Participant	School	Top 10% (≥178)		Top 5% (≥186)		Top 1% (≥195)	
		Cls.	School %ile	Cls.	School %ile	Cls.	School %ile
Oleksandr Volotko	Zaporizhzhia Gymnasium #93	5	91	2	87–91.4	0	0–93.4
Panel B: Female participants							
Yana Kolodach	Kyiv-Pechersk Lyceum #171 "Leader"	42	99.6	35	99.7	24	99.9
Olena Shulha	Cherkasy Physics-Math Lyceum (FIMLI)	56	99.8	49	99.9	20	99.9
Sofia Ryabchun	Rusanivskyi Lyceum	51	99.8	45	99.8	17	99.8
Kateryna Khomenko	Kherson Specialized School #30	49	99.7	35	99.7	13	99.8
Karolina Susol	Uzhhorod Specialized Boarding School	34	99.5	24	99.5	10	99.7
Anna Bondar	Lyceum #5 (Ivano-Frankivsk)	12	97	7	97.3	2	98.1
Tetiana Khomenko	Kremenchuk Lyceum #4 "Kremin"	32	99.4	14	99.1	1	93–97.5
Daria Zhelizna	Lyceum #15 (Chernihiv)	18	98.4	8	97.7	1	93–97.5
Anhelina Morhai	Mariupol City Lyceum	18	98.4	7	97.3	1	93–97.5
Panel C: Medians							
Male participants		*37.5*	*99.6*	*27.0*	*99.6*	*7.5*	*99.7*
Female participants		*34.0*	*99.5*	*24.0*	*99.5*	*10.0*	*99.7*

Notes: All 34 grade-11 participants in the 2021 UMO; 33 are shown, one having no identifiable school. The table follows Ellison and Swanson (2010)'s Table 2 in measuring school strength at three successively more demanding thresholds. "Cls." counts the participant's schoolmates scoring at or above the top decile, top twentieth, and top percentile of the 2021 ZNO math distribution — 178, 186, and 195 points, reached by 9.6, 5.0, and 0.9 percent of test-takers respectively. The corresponding school coverage is 3,395, 2,268, and 591 schools with at least one such scorer. All counts are leave-one-out, excluding the participant's own score where it clears the threshold. "School %ile" ranks that count against all 8,957 Ukrainian schools with a math test-taker in 2021. Percentiles are capped at 99.9, as in Table 5. Where a count ties a large block of schools — 62.1, 74.7, and 93.4 percent of schools have no qualifying student at the three cutoffs — the percentile is reported as the range spanning that block rather than a single number, matching Ellison and Swanson's treatment of the same case. Panel C reports the median leave-one-out count and the percentile that count occupies in the national school distribution, again following their median rows; it is the percentile of the median count, not the median of the percentiles. Every participant clears the first two cutoffs, but five score below 195, so at the top-percentile measure two participants at the same school can differ by one classmate, as Anton Uvarenko and Arsenii Liulka do at Lyceum #35. Participant names follow the transliteration used in Table 5 for the seven students appearing in both tables. † marks a school assignment that is inferred rather than confirmed. Source: UMO official results (uoi.ua), matched to UCEQA ZNO administrative microdata (testportal.gov.ua).

Appendix D Tail Gap Across Assessments

Table 7 assembles every Ukrainian assessment for which student-level microdata and a gender identifier are available, ordered by the age of the students tested. The purpose is to locate, approximately, the point in schooling at which the upper-tail gender gap becomes visible. Three features of the assembled evidence are worth stating before reading it. The grade-4 assessments show no detectable gap: MDYAPO's ratios sit slightly below parity and TIMSS 2007's slightly above, and every interval in both panels contains one. By grade 8 the gap is detectable in the more recent cycle, with TIMSS 2011 excluding parity at the two shallowest thresholds, though TIMSS 2007 at the same grade does not. By age 15 PISA excludes parity through the top 1 percent, and on the grade-11 ZNO the gap is precisely estimated at every depth and widens monotonically. The final panel adds the National Math Olympiad, whose competitors can be ranked past the point where the ZNO score distribution is exhausted; there the ratio continues to climb, reaching roughly ten to one among the deepest scorers.

The comparison is cross-sectional rather than a cohort panel, with one exception: TIMSS 2007 grade 4 and TIMSS 2011 grade 8 follow the same Ukrainian cohort four years apart, and are the only two panels that can be read as tracking the same students. The remaining panels compare different cohorts assessed by different instruments, so differences across panels confound age, cohort, and test design. The survey sources also differ enormously from ZNO in precision: PISA, TIMSS, and MDYAPO each rest on samples of three to six thousand students, so their deepest cells contain a few dozen students and their intervals are correspondingly wide, whereas ZNO is an administrative census of the full cohort. The top 0.1 percent is omitted throughout because it would leave three to six students in every survey panel.

Table 7: Upper-Tail Gender Gap Across Ukrainian Assessments, by Age of Students

Threshold	N	M:F Ratio	95% CI
Panel A: MDYAPO, grade 4 (2018); $N = 4{,}454$			
Top 10%	445	0.87	[0.68, 1.11]
Top 5%	223	0.88	[0.63, 1.22]
Top 1%	45	0.90	[0.44, 1.85]
Top 0.5%	22	0.60	[0.19, 1.86]
Panel B: TIMSS, grade 4 (2007); $N = 4{,}292$			
Top 10%	429	1.26	[0.99, 1.59]
Top 5%	215	1.37	[0.98, 1.92]
Top 1%	43	1.65	[0.72, 3.80]
Top 0.5%	21	1.96	[0.64, 6.00]
Panel C: TIMSS, grade 8 (2007); $N = 4{,}424$			
Top 10%	442	1.00	[0.80, 1.25]
Top 5%	221	1.10	[0.79, 1.54]
Top 1%	44	1.60	[0.59, 4.37]
Top 0.5%	22	1.67	[0.45, 6.18]
Panel D: TIMSS, grade 8 (2011); $N = 3{,}378$			
Top 10%	338	1.43	[1.10, 1.85]
Top 5%	169	1.57	[1.05, 2.37]
Top 1%	34	1.64	[0.66, 4.09]
Top 0.5%	17	1.55	[0.47, 5.12]
Panel E: PISA, age 15 (2018); $N = 5{,}998$			
Top 10%	600	1.47	[1.17, 1.84]
Top 5%	300	1.66	[1.23, 2.24]
Top 1%	60	2.25	[1.03, 4.92]
Top 0.5%	30	2.00	[0.66, 6.05]
Panel F: ZNO, grade 11 (2016–2021); $N = 643{,}977$			
Top 10%	64,214	1.32	[1.30, 1.34]
Top 5%	34,374	1.48	[1.45, 1.52]
Top 1%	6,327	2.10	[1.99, 2.22]
Top 0.5%	2,794	2.60	[2.39, 2.82]
Panel G: National Math Olympiad, grade 11 (2016–2021); $N = 207$			
Top 10%	21	9.50	[2.46, 36.71]
Top 5%	11	10.00	[1.65, 60.61]
Top 1%[a]	3	—	(0 female)

Notes: Panels are ordered by the age of the students assessed. The M:F ratio is males per female at or above each threshold, so a ratio of one indicates parity. Confidence intervals apply the Wilson score interval to the underlying female share and transform the bounds monotonically to the ratio scale, which is valid

Appendix E Sample Exam Items

Six hardest items on the 2021 ZNO math exam by P-value (Ukrainian Center for Educational Quality Assessment, 2021).

26. (P-value 7.8%) To prepare a disinfectant solution, concentrate is diluted with water in a mass ratio of 2:7, after which 1 g of fragrance is added for every 10 g of water. How many grams of concentrate are needed to prepare 485 g of solution?

27. (P-value 6.8%) Evaluate $\sqrt{9a^2 - 24a + 16} - \sqrt[3]{27a^3}$ for $a = 0.7$.

28. (P-value 10.6%) Solve the equation $x^4 - x^2 - 20 = 0$. Report the product of all its real roots.

32. (P-value 1.9%) The axial cross-section of a cylinder is rectangle $ABCD$, with side AD on the cylinder's lower base and diagonal AC meeting the base plane at angle β. Point K lies on the lower base's circle so that arc AK measures $90°$. Sketch the cylinder and identify the angle γ between plane (KBD) and the base plane, justifying its position, then find γ.

33. (P-value 10.3%) Prove the identity $\dfrac{2a^2 + 5a - 3}{a + 3} = \dfrac{1 - 2a}{2\cos 240°}$.

34. (P-value 2.5%) Given the system $ax^2 + 3ax + 4^{1+\sqrt{y}} = 8$, $x + 2 \cdot 4^{\sqrt{y}} = 1$, where x, y are variables and a is an arbitrary constant: solve the system when $a = 0$, then determine all solutions as a function of a.